

TOWARDS REVERSIBLE MODEL MERGING FOR LOW-RANK WEIGHTS

Anonymous authors

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ABSTRACT

Model merging seeks to replace multiple fine-tuned models with a more compact representation while retaining performance across the source tasks. Existing methods report competitive averages for full-rank fine-tuned models, but their quantitative behavior is much less favorable once the task updates are stored as low-rank weights, either through low-rank adaptation (LoRA) or post-training singular value decomposition (SVD). We first show empirically that standard single-model merging baselines suffer large average-score drops in this low-rank setting, with the degradation appearing consistently across ranks, compression types, and tasks. Motivated by this evidence, we propose to change the comparison target: rather than forcing all adapters into one set of weights, we store a compact basis (e.g., the storage equivalent of two or more models) from which each original task-specific model can be reconstructed by linear combination. This reframes merging as the construction of a reconstruction-capable model space, making it possible to “revert” to the appropriate task model when needed instead of relying on a single aggregate that cannot consistently match specialized models. We introduce Reversible Model Merging (RMM), an efficient, data-free, and flexible method with a closed-form solution for the basis weights and task-specific coefficients. Across datasets, model scales, compression ranks, and evaluation conditions, the reported results show that RMM delivers substantially higher average performance than TA, TIES, and DARE while preserving much of the accuracy of the low-rank compressed models at a controlled storage cost.

1 INTRODUCTION

Parameter-efficient fine-tuning methods, such as Low-Rank Adapters (LoRA) (Hu et al., 2021), make it practical to adapt large language models to many downstream tasks while training and storing only a small fraction of the full parameter set. Post-training singular value decomposition (SVD) truncation provides a complementary compression route, reducing storage and inference costs after standard fine-tuning (Ryu et al., 2023; Ping et al., 2024; Wang et al., 2025b;a). Model merging addresses the next scaling question: given several task-specific models, can one compact representation retain their aggregate task performance? Existing merging methods, from parameter interpolation to optimization-based procedures, show that full-rank fine-tuned models can often be consolidated with competitive average performance and without access to the original training data (Matena & Raffel, 2022; Yang et al., 2024; Yadav et al., 2023; Wei et al., 2025).

The empirical comparison becomes less favorable when the merged model is evaluated against the original task-specialized models rather than only against other merging baselines. Even strong merging strategies generally do not match the individual fine-tuned model on its own task, and the gap widens when the inputs to merging are low-rank LoRA or PT-SVD updates. A single merged parameter vector must average over distinct task directions; in practice, this comparison structure exposes task interference and measurable erosion of the adaptations that each source model learned (Gargiulo et al., 2025).

Low-rank adapters and SVD-truncated deltas are already compact at the single-task level, but the storage curve becomes unfavorable as the number of tasks grows. In multi-task, federated, or continual learning deployments, dozens or hundreds of task-specific adapters may accumulate, so the total memory cost again scales with the model count and the system must manage many separate

artifacts. The practical target is therefore not only high task accuracy, but also a favorable storage-versus-performance margin relative to storing every adapter independently. This motivates merging strategies that operate directly on low-rank representations, preserve the strongest empirical properties of the source models, and reduce redundancy without erasing task-specific signal.

We begin by quantifying where current approaches lose performance. The comparison is performed on task deltas, the weight differences between each fine-tuned model and the shared pre-trained initialization. Conventional methods combine these low-rank deltas into a single parameter set under an implicit linear-compatibility assumption. Our motivating experiments show that this assumption produces catastrophic average-score degradation for low-rank deltas: the baselines lose large margins relative to the unmerged compressed models and exhibit poor consistency across tasks. The pattern is explained by the limited expressive capacity of low-rank factors and by misalignment among independently optimized task subspaces, which interact destructively when forced into one update.

This empirical pattern suggests a different question: *What if the relevant quantitative target is not the best single merged model, but the best compact representation that can recover the task models?* Rather than collapsing all low-rank adapters into one set of weights, we keep a small shared basis whose linear combinations reconstruct the original adapters on demand. This design makes the storage budget explicit: retaining more than one basis component costs more than a single merged model, but it can recover much larger fractions of the task-specific performance. The resulting design space treats merging as construction of a reconstruction-capable model space, preserving complementary directions and enabling post-merging “reversion” to individual adapters.

We formalize this reconstruction-driven view as a basis-selection problem with a clear quantitative objective: minimize reconstruction error for the collection of task vectors while storing only a small number of basis directions and task-specific coefficients. This yields Reversible Model Merging (RMM), which has a closed-form solution for both the optimal basis and the coefficients used to recover each task adaptation. The empirical evaluation foregrounds the same comparison in every setting: RMM versus single-model baselines (TA, TIES, DARE), measured by task metrics, averaged performance, and relative storage. Across datasets and model configurations, RMM improves the average scores by large margins while retaining the efficiency benefits of low-rank parameterization, and the number of basis components gives a direct knob for the storage–performance trade-off.

2 RELATED WORK

Model merging provides the main baseline family for our comparison: it reduces the number of stored models, but its empirical value depends on how much task performance is preserved after consolidation. Existing methods broadly fall into two groups with different evidence and deployment profiles: data-free parameter-space manipulations, and optimization-based approaches that use data or auxiliary objectives to tune the merge. We review representative methods from both groups to clarify which storage, data, and performance trade-offs RMM is compared against.

Task Arithmetic: Task arithmetic (Ilharco et al., 2023) is the simplest data-free baseline in our experiments and therefore provides a useful lower-complexity comparison point. It defines each task delta as the difference between a fine-tuned model and the shared pre-trained initialization, then treats those deltas as modular directions in weight space. Adding, subtracting, or scaling deltas can transfer or blend behaviors, making the method attractive when storage must be reduced to one merged model. Its favorable simplicity, however, also exposes the central quantitative risk studied here: linear composition may preserve less task-specific performance when the deltas are low-rank.

DARE: DARE (Drop and Rescale) (Yu et al., 2024) targets the same interference that makes naive delta addition fragile. It discards a large fraction of parameter changes—up to 90-99%—and rescales the remaining updates so that the aggregate modification maintains its magnitude. This produces a single-model baseline with a very small storage footprint, and the pruning step is intended to keep the highest-impact task modifications while suppressing conflicting updates. In our framing, DARE is valuable precisely because it tests whether aggressive sparsification is enough to recover the performance lost by low-rank merging.

TIES: TIES-merging (Yadav et al., 2023) offers a more structured single-model comparison than randomized pruning. Its three stages—Trim, Elect Sign, and Merge—first retain large-magnitude

parameters, then enforce sign agreement across models, and finally average only the sign-consistent updates. This consensus mechanism is designed to improve stability by removing updates that would cancel or conflict. Related work such as EMR-Merging (Elect, Mask and Rescale) similarly addresses interference by electing a unified model and attaching task-specific binary masks and rescalers from individual models to steer the unified model toward each task (Huang et al., 2024). These methods sharpen the baseline comparison: they explicitly mitigate conflict, yet still ultimately rely on a single merged representation.

Other Merging Methods: Other merging approaches improve the performance side of the trade-off by using optimization or small amounts of data. Model soups (Wortsman et al., 2022) tune interpolation weights on validation loss, and Fisher-weighted averaging (Matena & Raffel, 2022) gives greater weight to parameters estimated to matter more for task performance. RegMean (Jin et al., 2023) and AdaMerging (Yang et al., 2024) adapt merging coefficients through objectives based on prediction discrepancies or unlabeled-data entropy, respectively. A more recent line minimizes the loss difference between the merged model and individual fine-tuned models using a first-order Taylor surrogate (Wei et al., 2025). These methods can improve empirical scores, but they change the evaluation condition by requiring labeled or unlabeled data, additional optimization, or both. RMM is positioned against the stricter data-free setting while targeting a larger practical performance margin than data-free single-model baselines provide for low-rank weights.

3 PRELIMINARIES

We first state the standard single-model objective that conventional baselines optimize, since this objective defines the quantitative comparison that later proves fragile under low-rank compression. Let $\{\theta_1, \dots, \theta_n\}$ denote n models fine-tuned on n distinct tasks. A merging algorithm $\mathcal{M}$ combines these models into one set of weights, $\theta_{\mathcal{M}} = \mathcal{M}(\{\theta_i\}_{i=1}^n)$. The usual target is to choose $\mathcal{M}$ so that this single model maximizes average validation performance over the n task metrics:

$$\max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\theta_{\mathcal{M}}, \mathcal{D}_i), \quad (1)$$

where $\mathcal{D}_i$ is the held-out validation set for task i , and Perf is the task’s evaluation metric (e.g., accuracy or F1 score). We assume, as is standard in merging, that every fine-tuned model starts from the same pretrained initialization θ_{pre} . For each layer l , the task-specific quantity being compared and merged is the update (or delta) $\Delta_i^l = \theta_i^l - \theta_{\text{pre}}^l$, where $\Delta_i^l \in \mathbb{R}^{m \times d}$ is the layer-wise weight difference for dimensions $m \times d$.

3.1 LOW-RANK COMPRESSED MODELS

In the setting studied here, the objects being compared are not full-rank deltas but low-rank compressed updates, chosen to reduce storage, training, and inference cost before any merging is attempted. We consider two common compression routes:

LoRA fine-tuning: During training, each layer receives trainable low-rank matrices $\mathbf{A} \in \mathbb{R}^{m \times r}$ and $\mathbf{B} \in \mathbb{R}^{r \times d}$, with $r \ll \min(m, d)$. The pretrained weights remain frozen, so the task-specific parameter budget is limited to $\mathbf{A}$ and $\mathbf{B}$.

Post-training SVD truncation (PT-SVD): After standard fine-tuning, the update Δ_i^l is decomposed by SVD and truncated to the top- r singular components. This yields the low-rank approximation $\hat{\Delta}_i^l = \mathbf{A}_i^l \mathbf{B}_i^l$, for $\mathbf{A}_i^l \in \mathbb{R}^{m \times r}$ and $\mathbf{B}_i^l \in \mathbb{R}^{r \times d}$, again with $r \ll \min(m, d)$.

Under either compression mechanism, model i contributes a layer-wise low-rank update $\hat{\Delta}_i^l = \mathbf{A}_i^l \mathbf{B}_i^l$, where $\mathbf{A}_i^l \in \mathbb{R}^{m \times r}$ and $\mathbf{B}_i^l \in \mathbb{R}^{r \times d}$, with r the compression rank. The low-rank merging problem is therefore to combine the compressed deltas $\hat{\Delta}_i = \{\hat{\Delta}_i^l\}_{l=1}^L$ while maintaining average task performance:

$$\hat{\theta}_{\mathcal{M}} = \theta_{\text{pre}} + \mathcal{M}(\{\hat{\Delta}_i\}_{i=1}^n), \quad \max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\hat{\theta}_{\mathcal{M}}, \mathcal{D}_i), \quad (2)$$

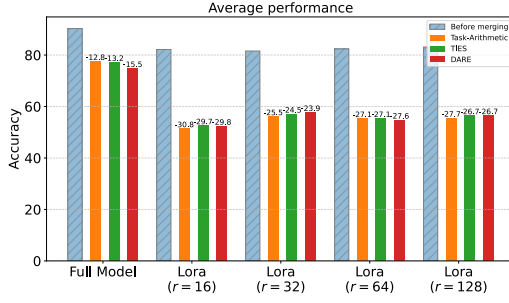


Figure 1: Average task performance comparison for LoRA models: unmerged individual models before merging versus single-model baselines and RMM across ranks $r \in \{16, 32, 64, 128\}$. The figure highlights the performance margin introduced by low-rank merging and the recovery obtained by reversible reconstruction.

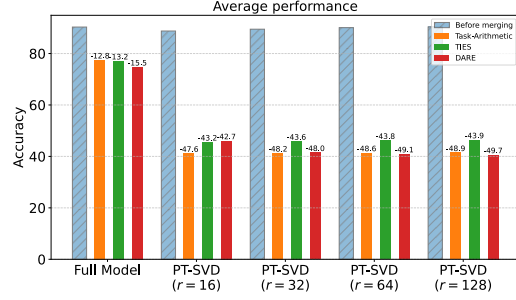


Figure 2: Average task performance comparison for PT-SVD models: unmerged individual models before merging versus single-model baselines and RMM across ranks $r \in \{16, 32, 64, 128\}$. The plotted averages show that the low-rank setting produces large baseline drops, while reversible reconstruction retains substantially more of the source-model performance.

where L denotes the total number of layers. All main comparisons in this work are conducted in this low-rank setting, so performance is always interpreted jointly with the compression rank, the number of tasks, and the storage ratio. The next section measures how conventional merging strategies behave under these conditions.

4 MOTIVATING OBSERVATION

We next make the low-rank comparison explicit by applying conventional merging strategies to the compressed factors introduced above. For each layer l , prior work has considered two natural extensions of merging (Tang et al., 2025; Zhao et al., 2025; Zheng et al., 2025; Zhang et al., 2025), which differ sharply in storage cost and therefore in practical relevance:

- **Combined merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{\mathbf{A}_i^l \mathbf{B}_i^l\}_{i=1}^n),$$

where each low-rank update is first reconstructed into full-rank form before the merge is applied.

- **Separate merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{\mathbf{A}_i^l\}_{i=1}^n) \mathcal{M}(\{\mathbf{B}_i^l\}_{i=1}^n),$$

where the two low-rank factors are merged independently and then recomposed.

Both formulations are valid algebraically, but their storage profiles differ by orders of magnitude. The combined strategy stores an $m \times d$ merged update, whereas the separate strategy keeps low-rank factors with the same scale as $\mathbf{A}_i^l \in \mathbb{R}^{m \times r}$ and $\mathbf{B}_i^l \in \mathbb{R}^{r \times d}$. Because $r(m + d) \ll md$ whenever $r \ll \min(m, d)$, the combined approach removes the very storage advantage that motivated compression. For example, with $m = d = 1024$ and $r = 16$, combined merging stores $md = 1,048,576$ parameters, while separate merging stores $r(m + d) = 16 \times (1024 + 1024) = 32,768$ parameters.

This is a **32 $\times$ storage reduction** for the separate representation, so the empirical comparison in the compressed regime should be made under separate merging. We therefore use the separate strategy as the practical single-model baseline condition for low-rank weights.

Using this storage-fair separate strategy, we evaluate several existing baselines on low-rank compressed RoBERTa-base models (Zhuang et al., 2021) across 5 GLUE tasks (Wang et al., 2018) (details in Appendix A.3). Figures 1 and 2 show the central empirical pattern: when the same baselines are applied to compressed models, the average task scores fall sharply relative to both unmerged compressed models and full-rank merging. The drop is consistent across ranks and compression mechanisms, indicating that low-rank merging changes the quantitative behavior of otherwise standard methods. We attribute this degradation to two factors: (i) **Amplified task interference:** compression discards much of the representation capacity, leaving low-rank models more susceptible to

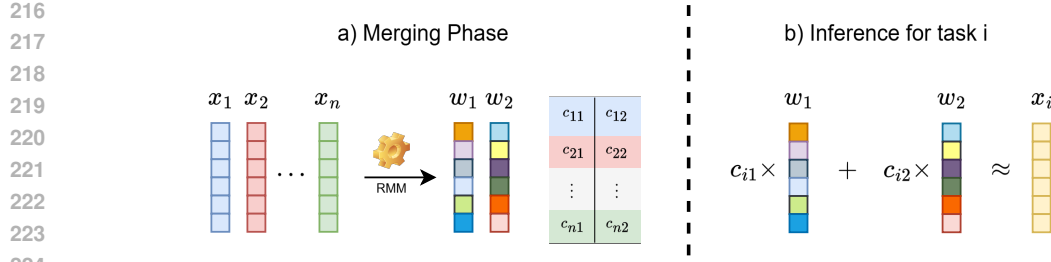


Figure 3: Overview of RMM as a storage–performance trade-off. **Left:** The merging phase replaces all task vectors with a compact set of basis directions (two in this example) plus task-specific coefficients. **Right:** At inference, model i is reconstructed from the same basis using its coefficient row, allowing evaluation against the corresponding task-specific model rather than a single aggregate.

destructive interference when merged; (ii) **Loss of coupling between A and B :** conventional merging treats low-rank matrices A_i^l and B_i^l as independent factors, ignoring the coupled nature of their product, leading to structural misalignment in the merged updates. Together, these factors explain why the single-model baselines lose large margins in the low-rank tables and why their task-level behavior is less consistent than the unmerged compressed models.

This evidence motivates the main design choice of the paper: if one merged model cannot reliably recover the task-specific scores of the original low-rank adapters, the storage budget should be spent on a compact representation that reconstructs those adapters more accurately. The relevant question becomes: *Can we approximate the original compressed models by retaining a small basis, equivalent to two or more adapters depending on the budget, and thereby recover a larger share of task performance?* Rather than forcing all updates into a single representation, we construct a *reconstruction-capable model space*. Next, we formalize this perspective and introduce *Reversible Model Merging* (RMM).

5 REVERSIBLE MODEL MERGING

The motivating results indicate that the limiting factor is not merely the choice of baseline, but the single-model target itself. A single aggregate has too little flexibility to preserve the task-specific scores of all low-rank compressed models. RMM therefore changes the target representation: it stores a small set of shared components, or basis models, from which each original task model can be approximately reconstructed and then evaluated under its own task condition.

Instead of forcing all n task-specific models into one merged solution $\hat{\theta}_{\mathcal{M}}$, RMM retains p merged components, where $2 \leq p \ll n$. These components span a low-dimensional subspace that captures the dominant variation among tasks, so increasing p gives the representation more capacity to recover the original adapters. The hyperparameter p is therefore directly interpretable in the experiments: it raises storage relative to a single merged model but improves reconstruction accuracy and downstream task scores.

For each layer l , RMM operates at the level of the r -dimensional vectors inside the low-rank factors A_i^l and B_i^l (r is the compression rank). Each row of A_i^l and each column of B_i^l is treated as an independent *task vector*, and a fixed row or column index is called a *task vector position*. At one such position, we collect the corresponding vectors across all n models and denote the vector from model i as $x_i \in \mathbb{R}^r$. The same reconstruction problem is then solved independently at every position: compress $\{x_i\}_{i=1}^n$ into a shared basis and later recover the vector required for each task. There are $m + d$ task vector positions per layer, consisting of m rows from A_i^l and d columns from B_i^l .

At a fixed task vector position, the compact representation consists of p basis vectors $\{w_1, \dots, w_p\}$, with each $w_j \in \mathbb{R}^r$ for $1 \leq j \leq p$. Their span, $W = [w_1 \dots w_p] \in \mathbb{R}^{r \times p}$, is selected to minimize reconstruction error for the n observed task vectors. The coefficient matrix $C \in \mathbb{R}^{n \times p}$ stores the comparison-specific weights: its i -th row $[c_{i1}, \dots, c_{ip}]$ reconstructs model i from the shared basis. The objective is therefore to choose W and C so that the aggregate reconstruction error over all n

task vectors is minimized:

$$\min_{\mathbf{W}, \mathbf{C}} \sum_{i=1}^n \left\| \mathbf{x}_i - \sum_{j=1}^p \mathbf{c}_{ij} \mathbf{w}_j \right\|_2^2. \quad (3)$$

This representation stores rp basis parameters and np coefficient parameters, for a total of $p(r+n)$ at each position instead of the rn parameters needed to retain all task vectors directly. The quantity p is the explicit storage–performance knob reported in the experiments: larger p increases storage but reduces reconstruction error. Figure 3 illustrates the $p = 2$ case. The optimization in equation 3 can be written in matrix form as:

$$\min_{\mathbf{W}, \mathbf{C}} \left\| \mathbf{X} - \mathbf{C} \mathbf{W}^\top \right\|_F^2, \quad (4)$$

where $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$ stacks the task vectors being reconstructed. For any fixed $\mathbf{W}$, the coefficient matrix with the smallest squared reconstruction error is the least-squares solution $\mathbf{C}^* = \mathbf{X} \mathbf{W} (\mathbf{W}^\top \mathbf{W})^{-1}$. With orthonormal basis vectors ($\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p, p < r$), the problem reduces to:

$$\min_{\mathbf{W}} \left\| \mathbf{X} - \mathbf{X} \mathbf{W} \mathbf{W}^\top \right\|_F^2 \quad \text{s.t.} \quad \mathbf{W}^\top \mathbf{W} = \mathbf{I}_p. \quad (5)$$

Thus the stored quantities are $\mathbf{W} \in \mathbb{R}^{r \times p}$ and $\mathbf{X} \mathbf{W} \in \mathbb{R}^{n \times p}$, where $\mathbf{X} \mathbf{W}$ is the coefficient matrix $\mathbf{C}$ from equation 4.

Theorem 1 (Optimal Solution to equation 5). *Let $\mathbf{X} \in \mathbb{R}^{n \times r}$ be zero-centered and $p < \min\{r, n\}$. The optimal basis $\mathbf{W}^*$ that minimizes equation 5 is given by the top- p eigenvectors of the sample covariance matrix $\mathbf{X}^\top \mathbf{X}$. Equivalently, $\mathbf{W}^*$ consists of the top- p right singular vectors of $\mathbf{X}$.*

The proof of Theorem 1 is provided in Appendix A.1. The theorem gives the closed-form basis used in all reported experiments, so the empirical comparisons do not require training data or iterative optimization. We next describe the two phases of RMM.

5.1 MERGING PHASE

The merging phase constructs the compact representation used for every later task comparison. For each layer and task vector position (a fixed row in $\mathbf{A}_i^l$ or column in $\mathbf{B}_i^l$), we gather the n corresponding task vectors as $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$. We compute the mean vector $\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ and zero-center the matrix (i.e., $\mathbf{X} \leftarrow [\mathbf{x}_1 - \boldsymbol{\mu} \dots \mathbf{x}_n - \boldsymbol{\mu}]^\top$). We then perform SVD on $\mathbf{X} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{V}^\top$ and select the top- p right singular vectors, $\mathbf{W}^* = \mathbf{V}_{[:,1:p]} \in \mathbb{R}^{r \times p}$, which minimize the reconstruction error in equation 5. Finally, we store the coefficient matrix $\mathbf{C}^* = \mathbf{X} \mathbf{W}^* \in \mathbb{R}^{n \times p}$, whose i -th row $[c_{i1}, \dots, c_{ip}]$ gives the linear-combination weights for reconstructing $\mathbf{x}_i$.

After this offline phase, the original task vectors $\{\mathbf{x}_i\}_{i=1}^n$ are discarded. Each position retains only $\mathbf{W}^*$, $\mathbf{C}^*$, and $\boldsymbol{\mu}$, reducing storage from rn parameters for all task vectors to $p(r+n)+r$ parameters. This count is the basis of the storage ratios reported in the tables, and it makes the cost of improving reconstruction accuracy through larger p explicit.

5.2 INFERENCE PHASE

At inference, the evaluation condition matches common model-merging practice: the target task is either specified by the user or selected by an oracle router that maps the input to a model index. Given target task i , RMM reconstructs the corresponding task vector $\hat{\mathbf{x}}_i$ from the stored tuple $(\mathbf{W}^*, \mathbf{C}^*, \boldsymbol{\mu})$ at each task vector position. The reconstructed rows and columns are then concatenated to recover $\mathbf{A}_i^l$ and $\mathbf{B}_i^l$ for the target model.

For each position, $\mathbf{W}^* \in \mathbb{R}^{r \times p}$ stores the shared orthonormal basis, $\mathbf{C}^* \in \mathbb{R}^{n \times p}$ stores the task-specific coefficient rows, and $\boldsymbol{\mu} \in \mathbb{R}^r$ stores the mean vector. Reconstruction for task i is the corresponding coefficient-weighted basis combination shifted by $\boldsymbol{\mu}$:

$$\hat{\mathbf{x}}_i = \sum_{j=1}^p c_{ij}^* \mathbf{w}_j^* + \boldsymbol{\mu} = \mathbf{C}_{[i,:]}^* \mathbf{W}^{*\top} + \boldsymbol{\mu}.$$

The vector $\hat{x}_i \in \mathbb{R}^r$ approximates the original task vector x_i that was removed from storage. Applying the same reconstruction independently to every row of A_i^l , every column of B_i^l , and every layer yields the reconstructed low-rank delta $\hat{\Delta}_i$ for task-specific model i . Algorithm 1 summarizes both phases. The experiments below evaluate whether this extra reversible structure translates into higher task metrics than single-model baselines at comparable reported storage budgets.

Algorithm 1 Reversible Model Merging (RMM)

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1: Merging Phase:
2: Require: Low-rank deltas  $A_i^l$  and  $B_i^l$ , compression rank  $r$ , number of basis vectors  $p$ 
3: for  $l = 1$  to  $L$  do
4:   for each task vector position (each row in  $A_i^l$  or each column in  $B_i^l$ ) do
5:     Gather task vectors  $x_i$  of each model  $i$  ( $1 \leq i \leq n$ )
6:      $X \leftarrow [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$ 
7:      $\mu \leftarrow \sum_{i=1}^n x_i \in \mathbb{R}^r$ 
8:      $X \leftarrow [x_1 - \mu \dots x_n - \mu]^\top$ 
9:      $U, \Sigma, V^\top \leftarrow \text{SVD}(X)$ 
10:     $W^* \leftarrow V_{[:,1:p]} \in \mathbb{R}^{r \times p}$ 
11:     $C^* \leftarrow XW^* \in \mathbb{R}^{n \times p}$ 
12:    Store  $(W^*, C^*, \mu)$  and discard task vectors  $\{x_1 \dots x_n\}$ 
13:   end for
14: end for
15: Inference Phase:
16: Require: target task  $i$ , stored representation  $(W^*, C^*, \mu)$  for each task vector position
17: for  $l = 1$  to  $L$  do
18:   for each task vector position do
19:      $\hat{x}_i = C_{[i,:]}^* W^{*\top} + \mu$ 
20:   end for
21:    $\hat{A}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all row positions})$ 
22:    $\hat{B}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all column positions})$ 
23: end for
24:  $\hat{\Delta}_i = \{\hat{A}_i^l \hat{B}_i^l\}_{l=1}^L$ 
25: Inference with the reconstructed model  $\theta_{pre} + \hat{\Delta}_i$ 

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6 EXPERIMENTS

We evaluate RMM by making the main quantitative comparison explicit in every setting: task-specific compressed models before merging, data-free single-model baselines after merging, and RMM with different numbers of basis components. The experiments vary compression type, rank, task set, and model architecture to measure whether RMM preserves more task performance at a reduced storage cost.

6.1 EXPERIMENTAL SETUP

The main experiments use eight RoBERTa-base models fine-tuned on eight tasks (Zhuang et al., 2021). Each model is compressed independently with PT-SVD or LoRA at ranks $r \in \{16, 32, 64, 128\}$, so each row block in the results compares methods under the same dataset, rank, and compression condition. We compare RMM against Task Arithmetic (TA) (Ilharco et al., 2023), TIES-merging (Yadav et al., 2023), and DARE (Yu et al., 2024); the unmerged individual compressed models are reported as the reference upper bound for the attainable task-specific score. Appendix A.2 extends the same comparison structure to OPT-1.3b (Zhang et al., 2022) and ViT-B/32 (Radford et al., 2021) with different task counts.

We follow the GLUE protocol (Wang et al., 2018) and report each task with its standard metric. QNLI (question–answering inference), MRPC (paraphrase detection), SST-2 (sentiment classification), MNLI (multi-genre natural language inference), QQP (duplicate question detection), and RTE (textual entailment) use accuracy; CoLA (linguistic acceptability) uses Matthews correlation; and

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STSB	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	77.98	58.05	90.59	83.80
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.19	31.66
	TIES	13%	50.56	34.78	49.08	31.82	36.82	47.29	0.00	3.75	31.76
	DARE	13%	50.80	33.51	49.08	31.82	36.82	47.29	0.00	5.07	31.80
	RMM ($p = 2$)	50%	51.20	78.20	88.30	66.74	86.02	47.29	6.56	33.59	57.24
	RMM ($p = 3$)	69%	88.83	86.26	92.89	79.58	87.18	48.74	33.85	60.46	72.22
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	78.34	59.33	90.68	84.46
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.21	31.66
	TIES	13%	50.56	35.19	49.08	31.82	36.82	47.29	0.00	3.87	31.83
	DARE	13%	50.72	33.51	49.08	31.82	36.82	47.29	0.00	3.48	31.59
	RMM ($p = 2$)	44%	50.54	74.32	55.05	62.68	87.07	47.29	0.00	23.18	50.02
	RMM ($p = 3$)	59%	84.66	86.32	92.89	79.70	88.05	47.29	23.57	44.54	68.38
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	78.70	61.11	90.72	85.10
	TA	13%	50.61	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.13	49.08	31.82	36.82	47.29	0.00	3.80	31.82
	DARE	13%	50.58	33.51	49.08	31.82	36.82	47.29	0.00	5.22	31.79
	RMM ($p = 2$)	41%	50.53	73.10	49.08	58.48	86.98	47.29	0.00	17.50	47.87
	RMM ($p = 3$)	55%	74.14	86.20	92.43	78.83	89.00	47.29	13.16	33.70	64.34
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	79.78	60.16	90.74	85.30
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.54	49.08	31.82	36.82	47.29	0.00	3.93	31.88
	DARE	13%	50.61	34.78	49.08	31.82	36.82	47.29	0.00	2.41	31.60
	RMM ($p = 2$)	39%	50.54	70.78	49.08	54.69	84.68	47.29	0.00	14.53	46.45
	RMM ($p = 3$)	52%	59.20	85.80	92.32	77.50	89.02	47.29	6.56	26.94	60.58

Table 1: GLUE performance for eight RoBERTa-base models compressed with PT-SVD. Each rank block compares unmerged compressed models (100% storage), single-model baselines (13% storage), and RMM at $p = 2$ or $p = 3$, with task metrics and the average score shown side by side.

STS-B (semantic textual similarity) uses Pearson correlation. The average over tasks is the primary summary statistic because it directly compares multi-task performance across methods. Storage is reported as the fraction of parameters retained relative to storing all individual task-specific adapters. Single-model baselines have fixed cost $\frac{r}{rn} = \frac{1}{n}$, while RMM with p basis components has relative cost $\frac{p(r+n)+r}{rn}$ as derived in Section 5.1.

6.2 RESULTS AND DISCUSSION

Tables 1 and 2 report the main comparison under matched ranks, tasks, metrics, and storage definitions. Across both PT-SVD and LoRA, and for every $r \in \{16, 32, 64, 128\}$, RMM achieves higher average scores than TA, TIES, and DARE. The margins are especially large at low ranks: at $r = 16$ and $r = 32$, the single-model baselines average only 31–42%, with particularly weak values on CoLA and STS-B. Under the same conditions, RMM with $p = 2$ reaches 57.24% for PT-SVD and 51.41% for LoRA at $r = 16$, and RMM with $p = 3$ further improves the averages to 72.22% and 58.64%. These scores do not fully match the unmerged compressed-model upper bounds, but they recover a substantially larger share of that performance than any single-model baseline.

The advantage remains visible at higher ranks, where the unmerged compressed models are stronger and the reconstruction problem is harder. With PT-SVD, RMM at $p = 3$ achieves 64.34% and 60.58% for $r = 64$ and $r = 128$, respectively; with LoRA, the corresponding averages are 55.40% and 61.35%. In the same rank blocks, TA, TIES, and DARE remain near 31–42% for PT-SVD and near 40% for LoRA, so the ordering is consistent across compression regimes. Even the lower-storage RMM variant with $p = 2$ preserves clear average-score gains, showing that the reversible basis captures task-specific signal that is lost when all updates are collapsed into one model.

The relative gain narrows as r increases, which is consistent with a higher-dimensional reconstruction problem: larger-rank task representations can express more task-specific variation and may be less aligned with a small shared basis. Nevertheless, the empirical pattern is stable across all reported ranks and both compression types: increasing storage from a single merged model to a reversible $p = 2$ or $p = 3$ representation buys substantially better task performance. The additional storage is offline memory, typically CPU-side and inexpensive relative to runtime quality, whereas the average-score improvements directly affect task outcomes. This makes RMM’s storage–performance trade-off practically favorable when accuracy is more valuable than the smallest possible artifact.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	68.23	53.92	88.71	77.71
	TA	13%	50.59	33.51	72.13	34.79	64.46	47.29	0.00	21.81	40.57
	TIES	13%	50.63	33.68	74.31	35.64	67.45	47.29	0.00	26.38	41.92
	DARE	13%	50.63	33.51	73.74	35.97	65.35	47.29	0.00	21.62	41.01
	RMM ($p = 2$)	50%	51.13	34.14	90.71	60.48	84.41	47.29	0.00	43.14	51.41
	RMM ($p = 3$)	69%	55.92	38.14	93.12	82.95	87.30	47.29	6.56	57.81	58.64
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	62.82	53.12	88.53	76.54
	TA	13%	50.91	68.06	54.70	33.77	65.50	55.96	0.00	-0.48	41.05
	TIES	13%	51.03	68.06	56.77	34.24	66.27	58.12	0.00	1.02	41.94
	DARE	13%	50.96	64.64	63.07	35.12	63.66	55.60	0.00	1.01	41.76
	RMM ($p = 2$)	44%	52.08	65.39	81.54	62.46	77.06	57.40	0.00	9.95	50.73
	RMM ($p = 3$)	59%	54.62	69.97	92.20	81.29	85.97	57.40	4.64	24.08	58.77
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	67.15	58.31	88.70	78.25
	TA	13%	54.75	68.52	77.29	37.24	36.82	47.29	0.00	-3.53	39.80
	TIES	13%	54.82	68.75	80.73	37.67	36.82	47.29	0.00	-2.48	40.45
	DARE	13%	53.71	68.58	78.56	38.33	36.82	47.29	0.00	-4.28	39.88
	RMM ($p = 2$)	41%	55.74	69.28	87.27	42.81	77.72	47.29	0.00	4.91	48.13
	RMM ($p = 3$)	55%	59.07	70.14	90.83	79.11	82.06	47.29	0.00	14.70	55.40
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	57.04	59.35	89.68	77.71
	TA	13%	51.49	69.22	54.47	37.16	63.27	47.29	0.00	-1.74	40.15
	TIES	13%	51.91	68.93	57.34	37.81	63.36	47.29	0.00	-0.25	40.80
	DARE	13%	51.97	68.99	58.60	35.64	63.33	47.29	0.00	-0.44	40.67
	RMM ($p = 2$)	39%	57.18	67.83	88.53	49.54	84.64	47.29	0.00	28.00	52.88
	RMM ($p = 3$)	52%	59.13	39.19	92.09	80.08	86.39	47.29	15.45	71.18	61.35

Table 2: GLUE performance for eight RoBERTa-base models compressed with LoRA. The table reports the same method, storage, task-metric, and average-score comparisons as Table 1, making the LoRA versus PT-SVD pattern directly comparable across ranks $r = 16, 32, 64, 128$.

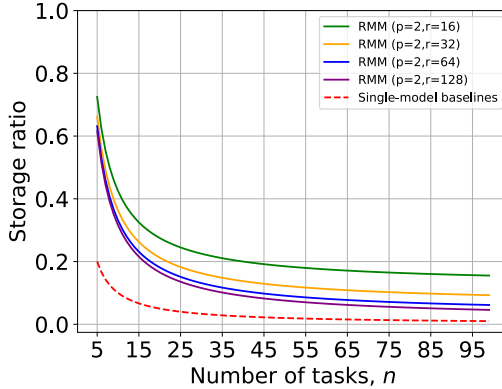


Figure 4: Storage scalability of RMM with $p = 2$. The plotted ratio is the percentage of memory required relative to storing all task-specific models, showing how the relative cost decreases as the number of tasks grows.

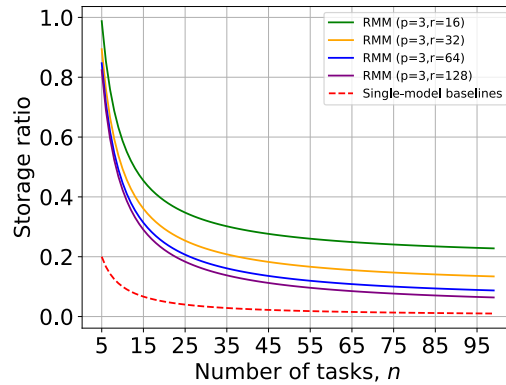


Figure 5: Storage scalability of RMM with $p = 3$. The curve uses the same relative-memory definition as Figure 4, highlighting the cost of a larger basis and its improved asymptotic behavior as more task models are included.

6.3 SCALABILITY OF RMM

We also evaluate the storage side of the trade-off as the number of models n increases. Figures 4 and 5 plot the storage ratio relative to storing all individual task-specific models for RMM with $p = 2$ and $p = 3$. In both cases, the relative cost decreases as more tasks are included, so the storage advantage improves in the regime where model collections are largest. This trend follows directly from the RMM parameter count: after the basis is stored, each additional task only adds a row to the coefficient matrix. Thus storage grows sublinearly relative to retaining every task-specific adapter. Conventional alternatives occupy the two extremes of the comparison: a single merged model has very low storage but loses substantial performance in the low-rank experiments, whereas storing every task-specific model preserves performance but scales linearly. RMM occupies the middle of this trade-off, making it suitable for large multi-task, continual, or federated settings

where many models must be managed without accepting the average-score losses observed for one-model merging.

7 CONCLUSION

This paper introduced Reversible Model Merging, a data-free framework that changes the quantitative target of low-rank model merging from one aggregate model to a compact reconstruction-capable representation. The evidence shows that conventional single-model baselines can be very storage-efficient but suffer large average-score drops when applied to LoRA or PT-SVD weights. RMM spends additional offline storage on a small basis and task-specific coefficients, and this consistently recovers substantially more of the individual compressed models’ task performance across the reported ranks, datasets, model families, and metrics. The hyperparameter p makes the trade-off explicit: larger bases require more storage but improve reconstruction fidelity and downstream scores.

Overall, the results support a more favorable framing for compressed-model deployment: practitioners need not choose only between a single low-performing merged model and linear growth from storing every adapter. By formalizing merging as a reversible reconstruction problem with a closed-form basis, RMM provides a middle point with clear empirical margins and interpretable storage accounting. This perspective opens future work on performance-aware compression, routing, and reconstruction strategies that report accuracy and memory jointly rather than treating merging quality as a single-model score alone.

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A APPENDIX

A.1 PROOF OF THEOREM 1

Let $\mathbf{A} = \mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top$. By definition of the Frobenius norm, $\|\mathbf{A}\|_F^2 = \text{Tr}(\mathbf{A}^\top \mathbf{A})$, where $\text{Tr}(\cdot)$ denotes trace of a square matrix. Expanding gives:

$$\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 = \text{Tr}((\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)).$$

We expand the term in $\text{Tr}(\cdot)$:

$$\begin{aligned} (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) &= (\mathbf{X}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top)(\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) \\ &= \mathbf{X}^\top \mathbf{X} - \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X} \\ &\quad + \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top. \end{aligned}$$

Next, we take the trace of the expanded form. Using cyclicity of trace and $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$:

$$\begin{aligned} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}). \end{aligned}$$

Since $\text{Tr}(\mathbf{X}^\top \mathbf{X})$ is a constant, minimizing $\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2$ is equivalent to maximizing $\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W})$. Hence,

$$\min_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 \iff \max_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}).$$

By the Rayleigh quotient theorem (Horn & Johnson, 2012), the maximizer is obtained by taking the columns of $\mathbf{W}$ to be the top- p eigenvectors of $\mathbf{X}^\top \mathbf{X}$. This is the principal component analysis (PCA) solution, which gives the p -dimensional subspace with the largest projected variance and, equivalently, the smallest squared reconstruction error for the stored task vectors. Moreover, if $\mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\top$ is the SVD of $\mathbf{X}$, then $\mathbf{X}^\top \mathbf{X} = \mathbf{V}\Sigma^2\mathbf{V}^\top$, so the eigenvectors of $\mathbf{X}^\top \mathbf{X}$ are precisely the right singular vectors (columns of $\mathbf{V}$). The same optimal basis is therefore obtained by selecting the top- p right singular vectors of $\mathbf{X}$, which is the implementation used to make the reconstruction objective numerically stable, especially when r is large.

A.2 ADDITIONAL RESULTS

This section extends the same comparison protocol beyond the main RoBERTa-base experiments. For **OPT-1.3b**, we evaluate six natural language understanding tasks from GLUE and related benchmarks: QNLI, MRPC, SST-2, MNLI, QQP, and BoolQ (Clark et al., 2019). For **ViT-B/32**, we evaluate seven vision classification datasets: MNIST (LeCun et al., 1998), GTSRB (Stallkamp et al., 2011), SVHN (Netzer et al., 2011), Cars (Krause et al., 2013), EuroSAT (Helber et al., 2019), DTD (Cimpoi et al., 2014), and SUN397 (Xiao et al., 2010). These additional settings test whether the RMM margin over single-model baselines persists across model families and modalities.

Table 3 reports six OPT-1.3b models compressed with PT-SVD. In every rank block, both RMM variants ($p = 2$ and $p = 3$) exceed the single-model baselines in average score, and $p = 3$ consistently moves closer to the unmerged compressed-model reference while using substantially less storage than retaining all models. Table 4 shows the same pattern for seven ViT-B/32 models. The vision results are particularly favorable: at $r = 128$, RMM with $p = 3$ reaches an average accuracy less than 3% below the unmerged models while using only 59% of the storage required to retain all models. These additional results show that the favorable RMM margins are not limited to the main GLUE setting.

A.3 RESULTS FOR 5 TASKS

For the motivating experiments in Section 4, RoBERTa-base is the pretrained model and the evaluation covers five GLUE tasks: QNLI, MRPC, SST-2, MNLI, and QQP. Figures 1 and 2 summarize the average-score comparison between unmerged compressed models, single-model baselines, and RMM. For completeness, Table 5 provides the task-level LoRA results corresponding to Figure 1, and Table 6 provides the task-level PT-SVD results corresponding to Figure 2.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	BOOLQ	Average
16	No merging	100%	91.74	71.54	95.30	84.19	88.14	71.47	83.73
	Task-Arithmetic	17%	50.34	37.16	58.49	34.43	63.30	41.87	47.60
	TIES	17%	49.79	36.75	61.24	35.57	63.55	40.67	47.93
	DARE	17%	49.95	37.68	56.08	34.51	63.39	42.78	47.40
	RMM ($p = 2$)	63%	88.12	44.00	93.58	73.06	85.63	59.69	74.01
	RMM ($p = 3$)	85%	91.58	49.04	95.64	80.31	87.52	65.50	78.27
32	No merging	100%	92.06	72.17	95.41	84.60	89.17	72.45	84.31
	Task-Arithmetic	17%	50.28	37.22	58.94	34.57	63.33	41.93	47.71
	TIES	17%	49.77	36.93	61.93	35.69	63.65	40.73	48.12
	DARE	17%	50.39	40.87	56.19	34.96	63.59	43.33	48.22
	RMM ($p = 2$)	56%	86.66	41.91	91.74	71.75	85.68	56.39	72.36
	RMM ($p = 3$)	76%	91.85	44.81	94.84	80.21	88.43	63.52	77.28
64	No merging	100%	92.26	72.12	95.41	84.64	90.07	72.84	84.56
	Task-Arithmetic	17%	50.28	37.28	59.06	34.56	63.34	42.08	47.77
	TIES	17%	49.75	37.04	61.81	35.84	63.74	40.76	48.16
	DARE	17%	50.16	40.52	58.83	35.59	63.35	42.54	48.50
	RMM ($p = 2$)	53%	83.87	42.72	89.45	69.76	85.70	53.43	70.82
	RMM ($p = 3$)	71%	92.07	41.97	94.50	79.70	88.73	61.96	76.49
128	No merging	100%	92.35	72.41	95.41	84.67	90.93	73.09	84.81
	Task-Arithmetic	17%	50.32	37.39	58.94	34.63	63.37	42.08	47.79
	TIES	17%	49.79	37.04	62.04	35.99	63.74	40.55	48.19
	DARE	17%	50.03	38.14	57.91	34.72	63.31	42.11	47.70
	RMM ($p = 2$)	52%	81.38	42.78	84.75	67.66	83.22	51.99	68.63
	RMM ($p = 3$)	69%	92.28	41.80	93.69	79.42	88.76	59.69	75.94

Table 3: GLUE and BoolQ performance for six OPT-1.3b models compressed with PT-SVD. Each rank block reports the storage ratio, per-task metrics, and average score for unmerged compressed models, single-model baselines, and RMM with $p = 2$ or $p = 3$.

r	Method	Storage	MNIST	GTSRB	SVHN	Cars	EuroSAT	DTD	SUN397	Average
16	No merging	100%	99.61	97.22	97.06	69.20	99.67	75.21	72.72	87.24
	Task-Arithmetic	14%	47.19	28.29	38.64	56.27	57.52	41.38	62.92	47.46
	TIES	14%	60.14	34.52	49.03	58.55	61.70	43.67	62.87	52.93
	DARE	14%	53.80	31.00	44.96	57.60	56.93	43.99	62.42	50.10
	RMM ($p = 2$)	55%	99.16	88.52	94.89	66.61	97.44	60.64	67.86	82.16
	RMM ($p = 3$)	76%	99.58	95.65	96.54	68.30	98.93	67.18	69.50	85.10
32	No merging	100%	99.65	98.45	97.35	73.04	99.78	78.72	75.52	88.93
	Task-Arithmetic	14%	47.90	28.31	38.68	56.24	57.52	41.38	63.00	47.58
	TIES	14%	61.25	35.28	49.95	58.80	62.41	44.04	63.05	53.54
	DARE	14%	53.05	31.04	46.34	58.54	57.74	42.87	62.87	50.35
	RMM ($p = 2$)	49%	98.98	88.23	94.29	68.32	96.81	61.91	68.36	82.41
	RMM ($p = 3$)	67%	99.50	96.53	96.68	70.90	98.93	70.43	71.23	86.31
64	No merging	100%	99.67	98.68	97.41	75.76	99.85	79.15	77.73	89.75
	Task-Arithmetic	14%	47.99	28.30	38.71	56.21	57.59	41.38	63.22	47.63
	TIES	14%	61.83	35.93	50.28	59.08	63.07	44.15	63.17	53.93
	DARE	14%	54.78	32.79	45.05	58.23	61.30	42.87	62.77	51.11
	RMM ($p = 2$)	46%	98.46	84.43	92.83	69.36	95.59	62.39	69.77	81.83
	RMM ($p = 3$)	62%	99.45	96.14	96.33	72.84	98.56	72.07	73.85	87.03
128	No merging	100%	99.68	98.75	97.46	76.99	99.85	79.04	78.59	90.05
	Task-Arithmetic	14%	48.00	28.31	38.73	56.21	57.63	41.44	63.35	47.67
	TIES	14%	62.08	36.36	50.48	59.23	63.41	44.68	63.25	54.21
	DARE	14%	55.16	32.07	45.47	58.55	61.85	43.09	62.85	51.29
	RMM ($p = 2$)	44%	97.22	78.56	90.63	70.22	94.00	61.49	71.03	80.45
	RMM ($p = 3$)	59%	99.28	94.59	95.95	74.24	98.19	71.86	76.02	87.16

Table 4: Vision classification performance for seven ViT-B/32 models compressed with PT-SVD. The table compares unmerged task-specific models, single-model merging baselines, and RMM variants by storage ratio, per-dataset accuracy, and average accuracy across ranks.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	82.16
	Task-Arithmetic	20%	50.61	33.51	73.85	34.23	64.58	51.36
	TIES	20%	50.65	33.51	75.57	35.83	66.97	52.51
	DARE	20%	50.61	33.51	79.36	34.70	63.40	52.32
	RMM ($p = 2$)	73%	59.64	68.64	93.00	82.68	86.89	78.17
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	81.57
	Task-Arithmetic	20%	51.18	67.30	58.83	34.33	68.56	56.04
	TIES	20%	51.36	68.17	59.52	34.76	71.46	57.05
	DARE	20%	50.83	64.87	68.12	35.54	68.93	57.66
	RMM ($p = 2$)	66%	57.72	67.54	92.55	81.80	86.13	77.15
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	82.37
	Task-Arithmetic	20%	54.74	68.93	80.28	35.70	36.82	55.29
	TIES	20%	55.26	68.93	78.78	36.76	36.82	55.31
	DARE	20%	54.71	68.81	74.54	38.72	36.82	54.72
	RMM ($p = 2$)	63%	59.82	67.48	91.51	83.21	84.41	77.29
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	83.12
	Task-Arithmetic	20%	52.52	68.70	54.82	37.83	63.46	55.47
	TIES	20%	52.55	69.04	57.80	38.91	63.72	56.40
	DARE	20%	55.04	69.62	54.47	38.92	64.12	56.43
	RMM ($p = 2$)	62%	60.21	34.96	91.86	78.59	86.65	70.45

Table 5: Five-task GLUE performance for RoBERTa-base models fine-tuned with LoRA. The table expands Figure 1 by showing storage ratio, per-task scores, and average score for unmerged models, single-model baselines, and RMM across ranks.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	88.76
	Task-Arithmetic	20%	53.03	35.19	49.08	31.82	36.82	41.19
	TIES	20%	55.46	54.32	49.08	31.82	37.21	45.58
	DARE	20%	53.69	58.67	49.08	31.82	37.08	46.07
	RMM ($p = 2$)	73%	79.72	85.16	90.83	76.58	86.09	83.68
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	89.46
	Task-Arithmetic	20%	53.23	35.59	49.08	31.82	36.82	41.31
	TIES	20%	55.78	55.19	49.08	31.82	37.62	45.90
	DARE	20%	55.89	33.74	49.20	31.82	36.85	41.50
	RMM ($p = 2$)	66%	63.87	82.78	91.28	77.12	87.05	80.42
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	90.05
	Task-Arithmetic	20%	53.49	35.88	49.08	31.82	36.82	41.42
	TIES	20%	56.14	55.94	49.08	31.82	38.08	46.21
	DARE	20%	53.51	33.68	49.08	31.82	36.82	40.98
	RMM ($p = 2$)	63%	56.12	81.22	91.06	76.74	88.07	78.64
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	90.34
	Task-Arithmetic	20%	53.60	36.06	49.08	31.82	36.82	41.48
	TIES	20%	56.40	56.35	49.08	31.82	38.61	46.45
	DARE	20%	50.52	34.96	49.08	31.82	36.92	40.66
	RMM ($p = 2$)	62%	54.11	79.19	90.60	74.49	87.46	77.17

Table 6: Five-task GLUE performance for RoBERTa-base models compressed with PT-SVD. The table expands Figure 2 with the task-level scores and storage ratios used to compare unmerged models, single-model baselines, and RMM.